CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

M. Tech. Mechanical (CAD/CAM) **Sem-I University Examination, 2010-11**

ME-704 ADVANCED MACHINE DESIGN

Date: 06/01/2011, Thursday **Time: 10:30 a.m. to 1:30 pm** **Maximum Marks:70**

Instructions:

1. Attempt **all** the questions.
2. Figures to the right indicate ***full*** marks.
3. Make suitable assumptions and draw neat figures wherever if required.
4. PSG Design data book is allowed in examination.

SECTION-I

Q1 (a) Discuss on various phases for a typical design process. **[05]**

Q1 (b) An alloy steel shaft made from material which has 6_{ut}=700Mpa and its endurance limit is 0.5 6_{ut} .it is subjected to a fluctuating stresses as shown below. **[08]**

6_m(MPa)	6_a(MPa)	% time
400	300	20
350	250	40
450	500	40

Estimate the life of shaft. Take k_a=0.85, k_b=0.8.

Q2 (a) Discuss on design for rigidity. How rigidity of mechanical system and Components can be increased. **[03]**

Q2 (b) For S-Gearing determine the dimensions of gear keeping ration in center distance constant m=7, α=20, i=8.5, a=182 mm, determine the diameter and contact ratio. **[09]**

OR

Q2 (a) Turbine shaft translates 700 KW at 800 rpm. The permissible shear stress is 90 N/mm2.while the twist is limited to 0.5 degree in a length of 2.5m.calculate the Diameter of shaft. Take G=0.8*105 N/mm2, if the shaft is chosen hollow with di/do=0.65.calculate percentage saving in material. **[06]**

Q2 (b) Discuss on creep phenomena with necessary curve. **[03]**

Q2 (c) Explain following materials for application at elevated temperature. **[03]**
1. Ferrite steel, 2.stainless steel, 3. Cobalt and nickel base alloys,

Q3 (a) An edge crack beam, there a crack in its central plane whose length is 5mm, a load of 1000N is applied opposite to crack. So that crack would tend to open in bending. Calculate the SIF of crack if the beam has the following dimensions. Depth of the beam w=25mm, thickness b=10mm.span S=100 mm. **[06]**

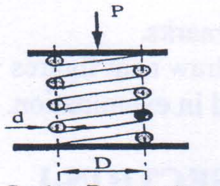
Q3 (b) Explain design consideration to be made for assembly. **[04]**

OR

Q3 (a)	Comparison of concurrent engineering and traditional engineering.	[03]
Q3 (b)	Explain system reliability for series, parallel and combined series parallel systems.	[03]
Q3 (c)	Explain Design for reliability.	[04]

SECTION-II

Q. 4(a) Prove that for a given helical spring, minimum weight for given conditions occurs when the spring is so designed that the maximum load on it is equal to twice the initial load. **[11]**



Spring Parameters

Q. 4(b) Explain (i) Normal, (ii) Redundant and (iii) Incompatible specifications **[03]**

Q. 5(a) How will you specify the E.O.T crane as per BIS? **[05]**

Q. 5(b) How will you classify the different material handling equipment? **[05]**

OR

Q. 5 Design the main girder for an over head traveling carne of following data: **[10]**

Lifting capacity=50000 N
Span = 15 mm
Assume weight of trolley = 22000 N
Young's modules = 2.1 X 10^5 N/mm^2
Trolley wheel base = 950 mm
Upper portion of truss = 153.6 mm
Deflection = 9 mm

Q6 (a) Discuss various quadrants of legal and ethical domain. **[06]**

Q6 (b) Explain codes of ethics. **[05]**

OR

Q6 (a) Discuss Human factor consideration for design of **[03]**
1) Lever of screw jack, 2) Clutch lever of an automobile car.
3) Driver's seat in automobile truck.

Q6 (b) Boron fibers (40 v %) are used to reinforce aluminum for an engineering application. Use the table provided below to calculate the density, modulus of elasticity, and tensile strength long the fiber orientation. Also estimate the modulus of elasticity if the load is to be applied perpendicular to the fibers. **[08]**

Materials	Density (g/cm^3)	Modulus (GPa)	Strength (MPa)
Boron Fiber	2.36	55	4000
Aluminum	2.70	10	50